PROJECT PROPOSAL – PROPOSED SOLUTION

Date	01 October 2026
Team ID	SWTID-2026-8237
Project Title	Comic Craft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Project Overview

Objective: To develop a web-based AI application that transforms user-provided story ideas into personalized comic stories, including structured panel outlines, narration, character dialogues, comic-style illustrations, and a downloadable PDF.

Scope: Comic Craft covers the complete comic-generation workflow from user prompt collection through AI-based story generation, image generation, panel layout, preview, and PDF export. The system is designed for non-artists, storytellers, hobbyists, and casual users.

Problem Statement

Description: Creating a complete comic traditionally requires separate skills for story writing, dialogue creation, illustration, panel arrangement, and document preparation. This can make comic creation time-consuming and difficult for users without artistic or technical expertise.

Impact: An automated solution can reduce the effort required to move from a simple creative idea to a complete comic. Users can customize the character, setting, tone, and art style while receiving a cohesive panel-by-panel story and illustrations in a single workflow.

Proposed Solution

Approach: Comic Craft combines Fast API, Google Gemini models, and Stable Diffusion to automate comic creation. Gemini Flash generates a structured 5-panel outline; Gemini Pro expands the outline into detailed narration and character dialogue; Stable Diffusion generates illustrations for each panel. The backend then builds the comic layout and exports the result as a PDF.

Key Features:

User input for story prompt, main character name, setting, story tone, and art style.

Structured 5-panel comic outline generation using Gemini Flash.

Detailed narration and character dialogue generation using Gemini Pro.

Comic-style illustration generation using Stable Diffusion and Hugging Face Diffusers.

Panel layout construction that associates each image with its corresponding story content.

Web-based comic preview using HTML, CSS, and Jinja2 templates.

PDF export using FPDF for saving, sharing, and offline viewing.

FastAPI routes for browser-based generation and JSON API-based generation.

Project Initialization and Planning Phase

Project Proposal (Proposed Solution) report – The proposal focuses on simplifying comic creation through an end-to-end generative AI pipeline. The system addresses the effort involved in writing stories, creating dialogues, producing illustrations, organizing panels, and exporting the final comic. The planned solution uses a modular architecture with a Fast API backend, Jinja2 frontend, Gemini models for text generation, Stable Diffusion for images, and FPDF for PDF generation.

Project Workflow

Step	Activity	Description
1	Collect user requirements	Capture story prompt, character, setting, tone, and art style.
2	Generate comic outline	Gemini Flash creates a structured 5-panel outline.
3	Generate story	Gemini Pro creates narration and character dialogues.
4	Generate illustrations	Stable Diffusion creates an image for each panel.
5	Build comic layout	Match panel images with titles, descriptions, captions, and narration.
6	Preview comic	Render the generated comic through Jinja2 templates.
7	Export PDF	Compile the complete comic into a downloadable multi-page PDF.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing resources	A system capable of running the Fast API application and AI/image-generation workflow; GPU support may be used for Stable Diffusion.
Software	Framework	Fast API, Unicorn
Software	Frontend	HTML, CSS, Jinja2
Software	AI / ML	Google Gemini API, Hugging Face Diffusers, Stable Diffusion, PyTorch

Software	PDF generation	FPDF
Software	Development	Python, pip, virtual environment, Git
Data / Inputs	User-provided creative inputs	Story prompt, character name, setting, tone, and art style
APIs / Services	Generative AI access	Gemini API and Hugging Face / Diffusers-based image generation

Expected Outcome

The completed Comic Craft system is expected to provide an end-to-end, user-friendly comic creation experience. A user can submit a creative prompt, receive a personalized multi-panel comic with generated story and illustrations, preview the result in the browser, and export the final comic as a PDF.

Source Basis

This document is prepared from the supplied Comic Craft project description and the supplied Proposed Solution template. The Comic Craft source specifies the application workflow, AI models, architecture, routes, frontend templates, image generation, layout construction, and PDF export.